The Heart.
Position & External Features

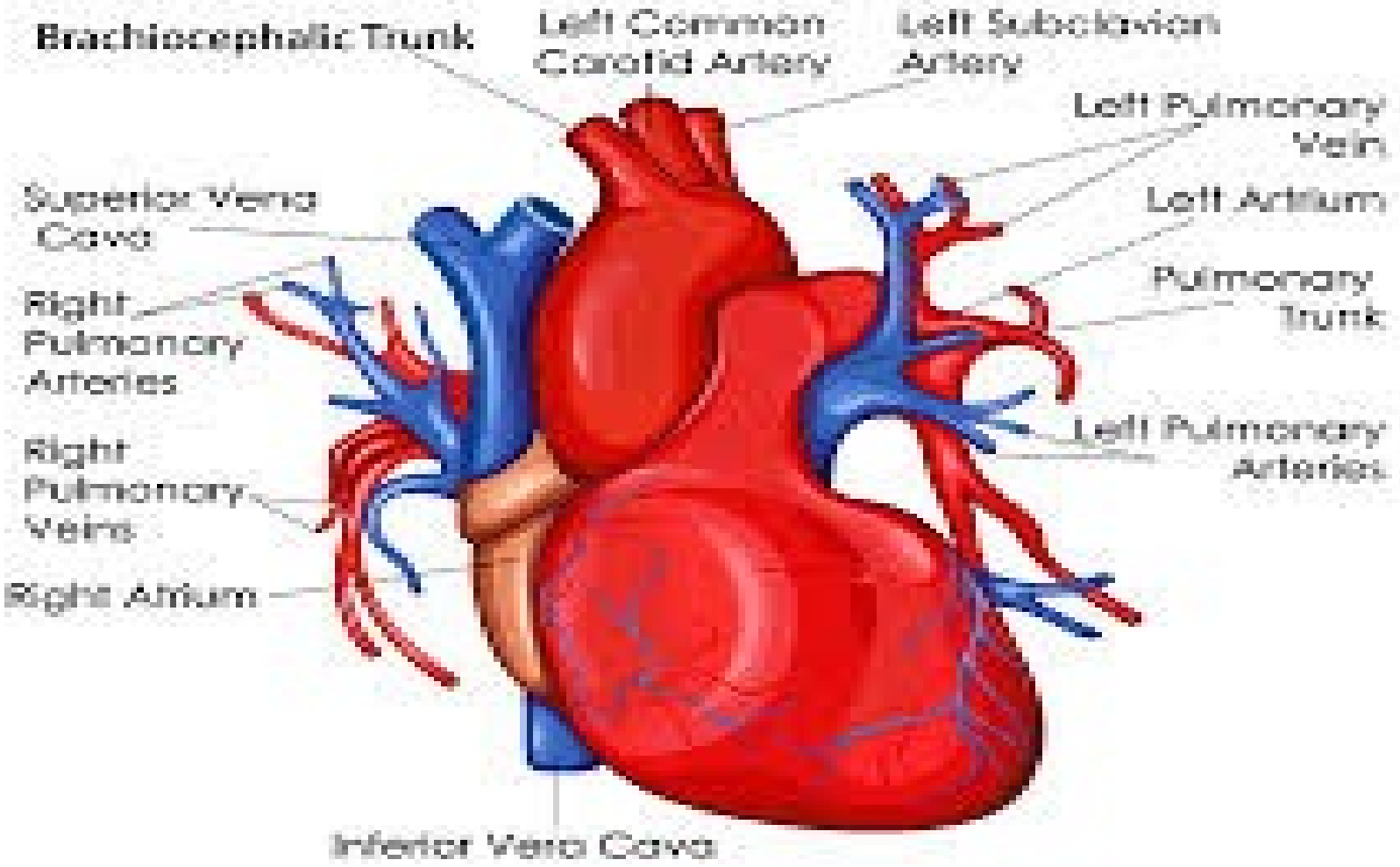
BY

DR AFSHEEN KHAN

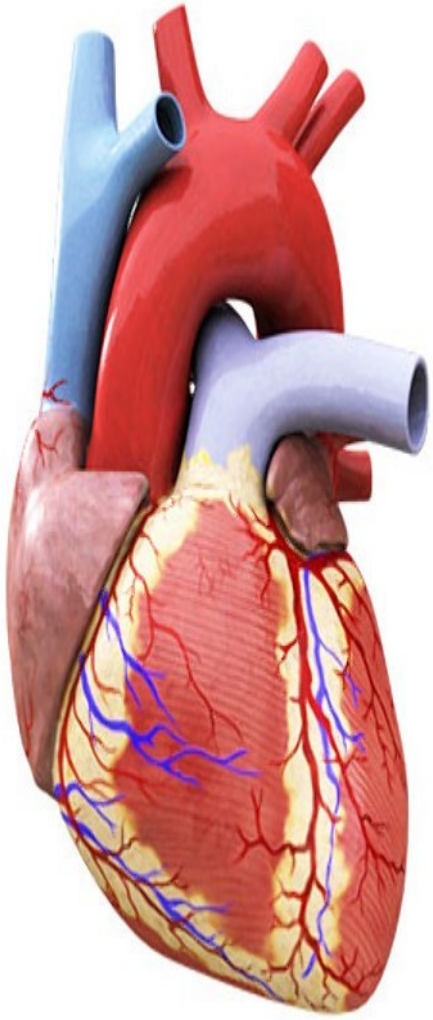
Learning Objectives:

- At the end of the demonstration, the student should be able to :
- Describe the **anatomical position** of the heart.
- Describe the **layers of the heart walls** and surrounding **pericardium.**
- Identify and describe the **chambers** and **valves** of the heart.
- Identify the **major blood vessels** to and from the heart.
- Describe the **blood flow** though the heart.

Heart Diagram



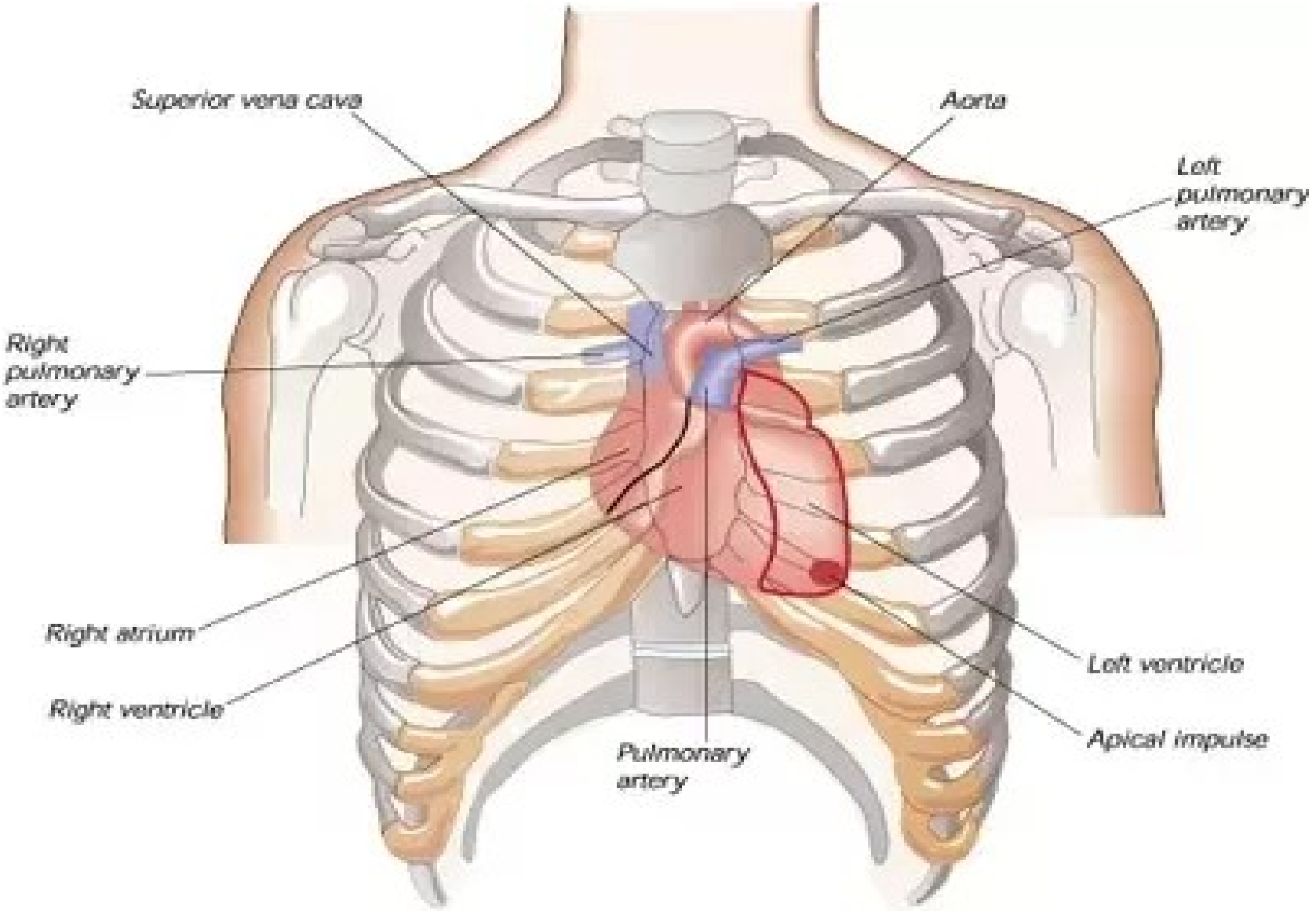
- Heart is a hollow organ that is pyramidal in shape.
- It lies in the pericardium in the middle mediastinum.
- It is connected at its base to the great blood vessels.



GENERAL FEATURES OF THE HEART

- The Heart Has
- An Apex And Base
- 2 Surfaces
- Sternocostal Surface
- Diaphragmatic Surface

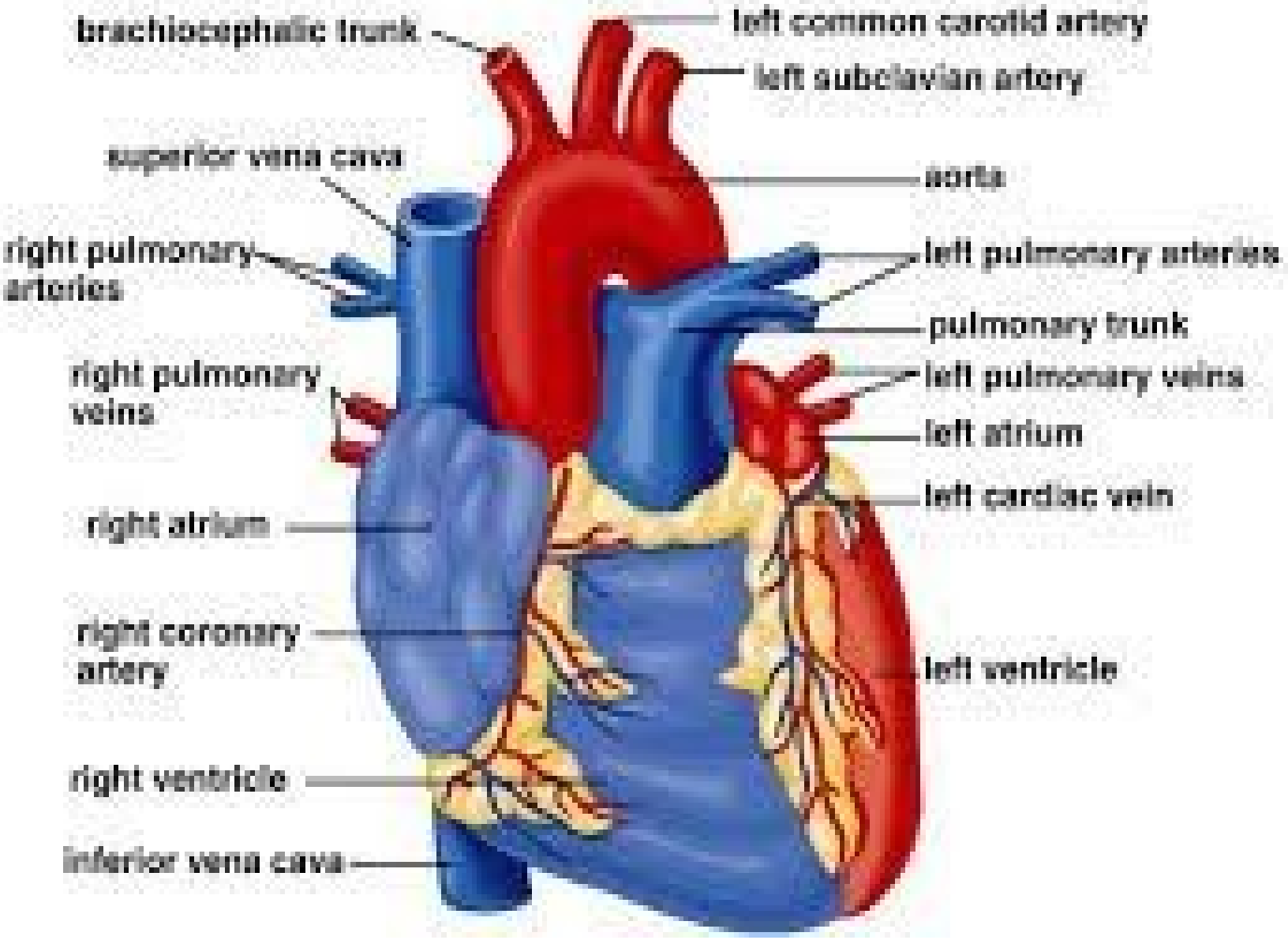
- 4 Borders
- Upper And Lower Border
- Right And Left Border


Superior vena cava
Aorta
Left pulmonary artery
Right pulmonary artery
Right atrium
Right ventricle
Pulmonary artery
Left ventricle
Apical impulse

THE APEX AND BASE OF THE HEART

- The Apex Of The Heart
- Formed By Left Ventricle,is Directed Downward,forward And To The Left
- It Lies At The Level Of The 5th Intercostal Space.9cm From The Midline.
- Base(posterior Surface)
- Formed By Left Atrium , Into Which Open The Four Pulmonary Veins.

External Heart Anatomy



POSITION

- The Heart Is Located Directly On **Top Of The Diaphragm Behind The Sternum**.
- It Is Positioned In The **Middle Mediastinum, Between The Left And Right Lungs.**

Position:

- **Right Border** Consists Entirely Of The Right Atrium.
- **Inferior Border** Is Made Up Mostly Of Right Ventricle With A Small Portion Of Left Ventricle.
- **Left Border** Is Mostly Left Ventricle, Auricle Of Left Atrium Forming Uppermost Part.

- **Anterior Or Sternocostal Surface:**
- Consists Of Right Atrium ,
- Vertical Atrioventricular Groove(coronary SULCUS),
- Right Ventricle With A Narrow Strip Of Left Ventricle.
- **Inferior Or Diaphragmatic Surface Consists:**
- Right Atrium Receiving Inferior Vena Cava, Anteroposterior Atrioventricular Groove
- **The Posterior Surface (Or Base) Consists Of:**
- Left Atrium, Receiving The Four Pulmonary Veins

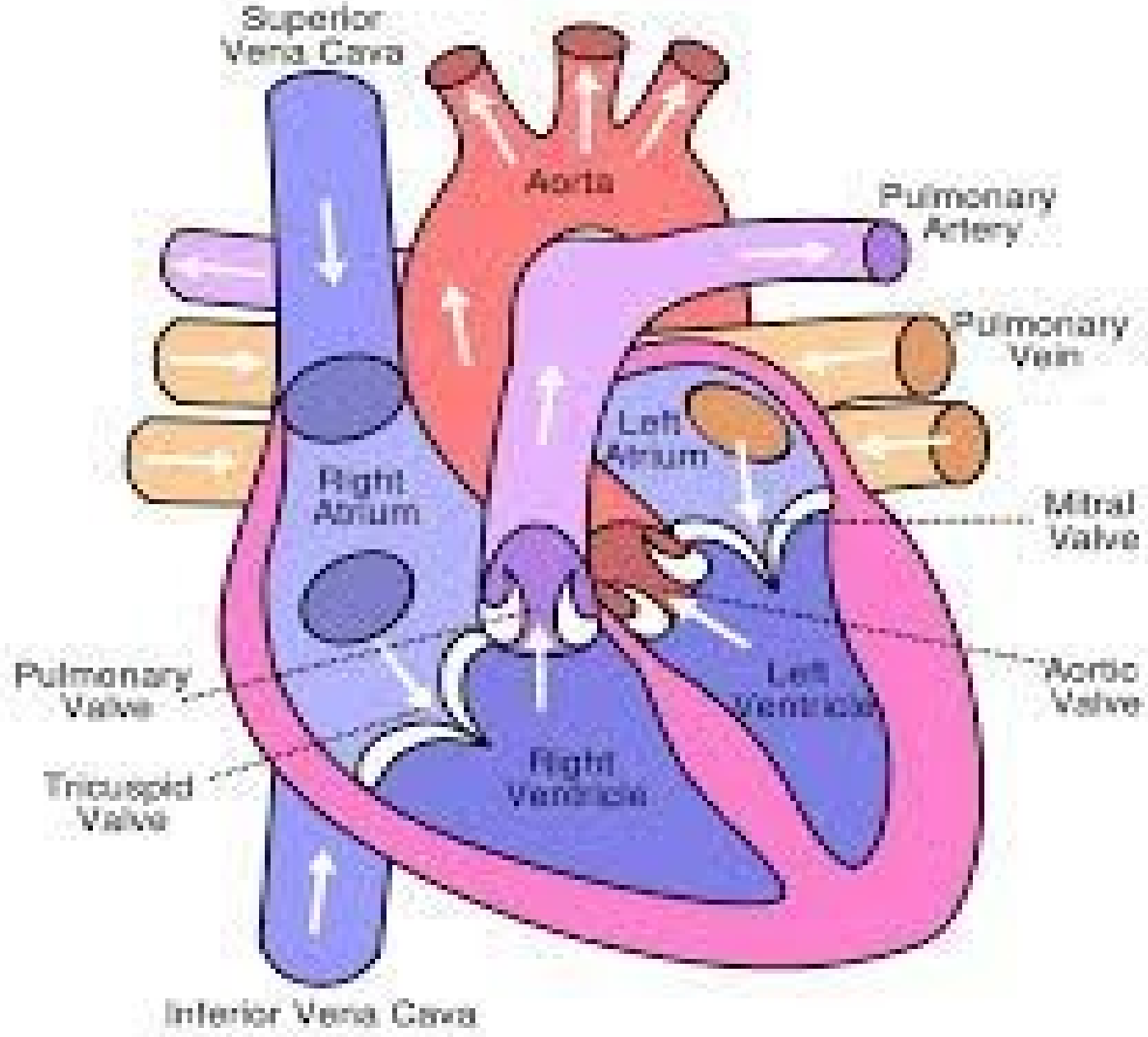
Anterior Or Sternocostal Surface

Consists Of Right Atrium ,

Vertical Atrioventricular Groove,

The Ventricular Part Divided Into Right Ventricle 2/3 Anf Left Ventricle 1/3 By The Anterior Interventricular Groove.

Right Ventricle With A Narrow Strip Of Left Ventricle

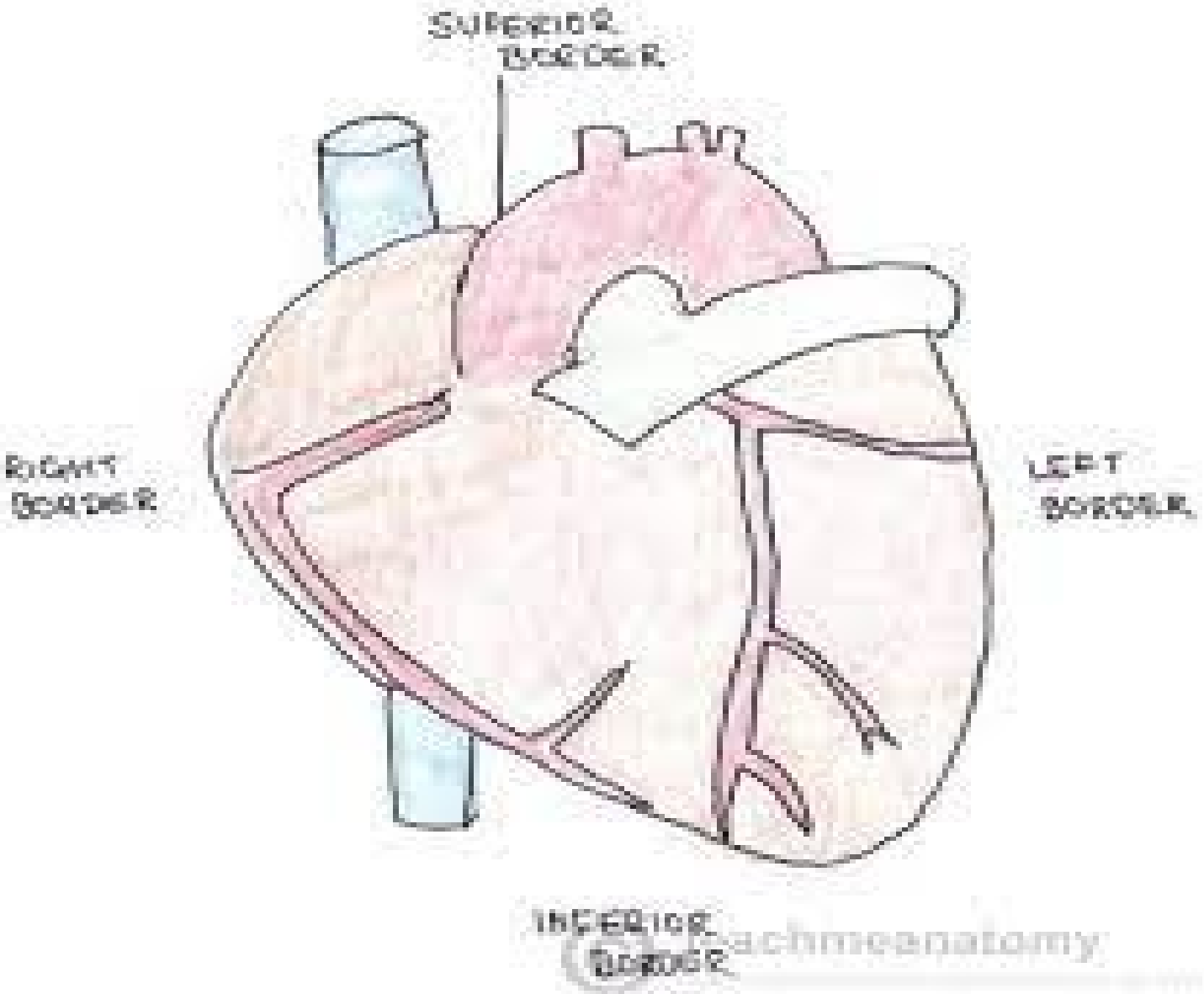

Superior Vena Cava
Aorta
Pulmonary Artery
Pulmonary Vein
Left Atrium
Right Atrium
Mitral Valve
Aortic Valve
Pulmonary Valve
Left Ventricle
Right Ventricle
Tricuspid Valve
Inferior Vena Cava

Inferior or Diaphragmatic surface consists:

This Surface Is Formed Mainly By The Right And Left Ventricles Seprated By The Posterior Interventricular Groove.

The Inferior Surface Of The Right Atrium,also Forms Part Of This Surface.

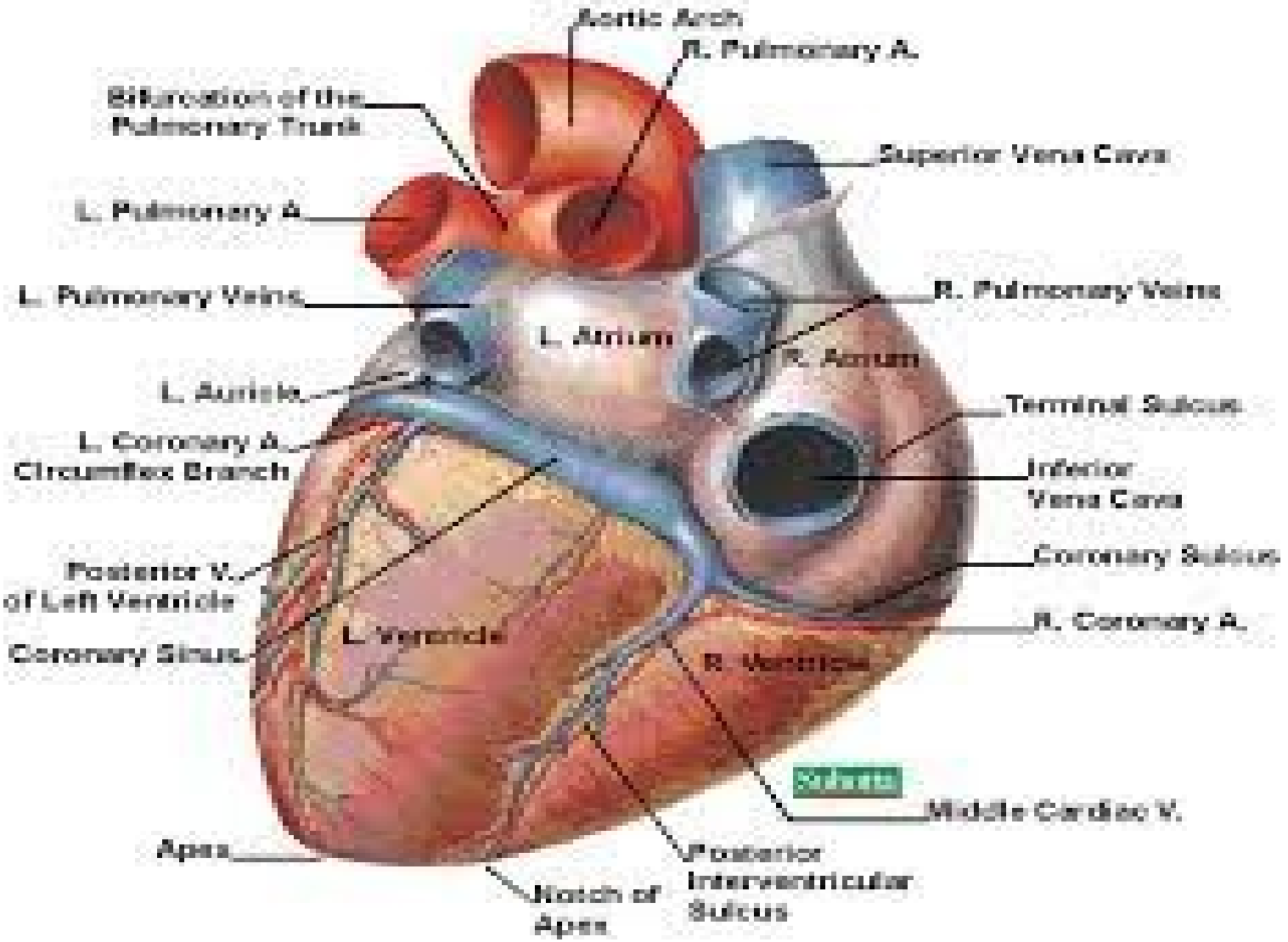
Right Atrium Receiving Inferior Vena Cava, Anteroposterior Atrioventricular Groove


SUPERIOR BORDER
RIGHT BORDER
LEFT BORDER
INFERIOR BORDER

The posterior surface (or base) consists of:

Left Atrium, Receiving The Four Pulmonary Veins

Position Varies **A Little** Between Systole And Diastole.


Aortic Arch
R. Pulmonary A.
Bifurcation of the Pulmonary Trunk
Superior Vena Cava
L. Pulmonary A.
L. Pulmonary Veins
R. Pulmonary Veins
L. Atrium
R. Atrium
L. Auricle
Terminal Sulcus
L. Coronary A. Circumflex Branch
Inferior Vena Cava
Posterior V. of Left Ventricle
Coronary Sulcus
Coronary Sinus
L. Ventricle
R. Coronary A.
R. Ventricle
Middle Cardiac V.
Apex
Posterior Interventricular Sulcus
Notch of Apex

BORDERS OF THE HEART

- THE RIGHT BORDER IS FORMED BY THE RIGHT ATRIUM
- LEFT BORDER BY THE LEFT VENTRICLE AND PART OF THE LEFT AURICLE
- LOWER BORDER FORMED MAINLY BY THE RIGHT VENTRICLE AND SLIGHTLY BY LEFT VENTRICLE
- SUPERIOR BORDER FORMED BY THE RIGHT AND LEFT ATRIA AND AURICLES IN AN ANTERIOR VIEW

- Roots of great vessels fix it, but the ventricles are free to move within the pericardium.
- **In full inspiration**, the apex of the heart descends more than the relatively fixed base, and heart occupies somewhat more vertical position.
- **In full expiration**, the ascent of the diaphgram forces the heart into more horizontal position.

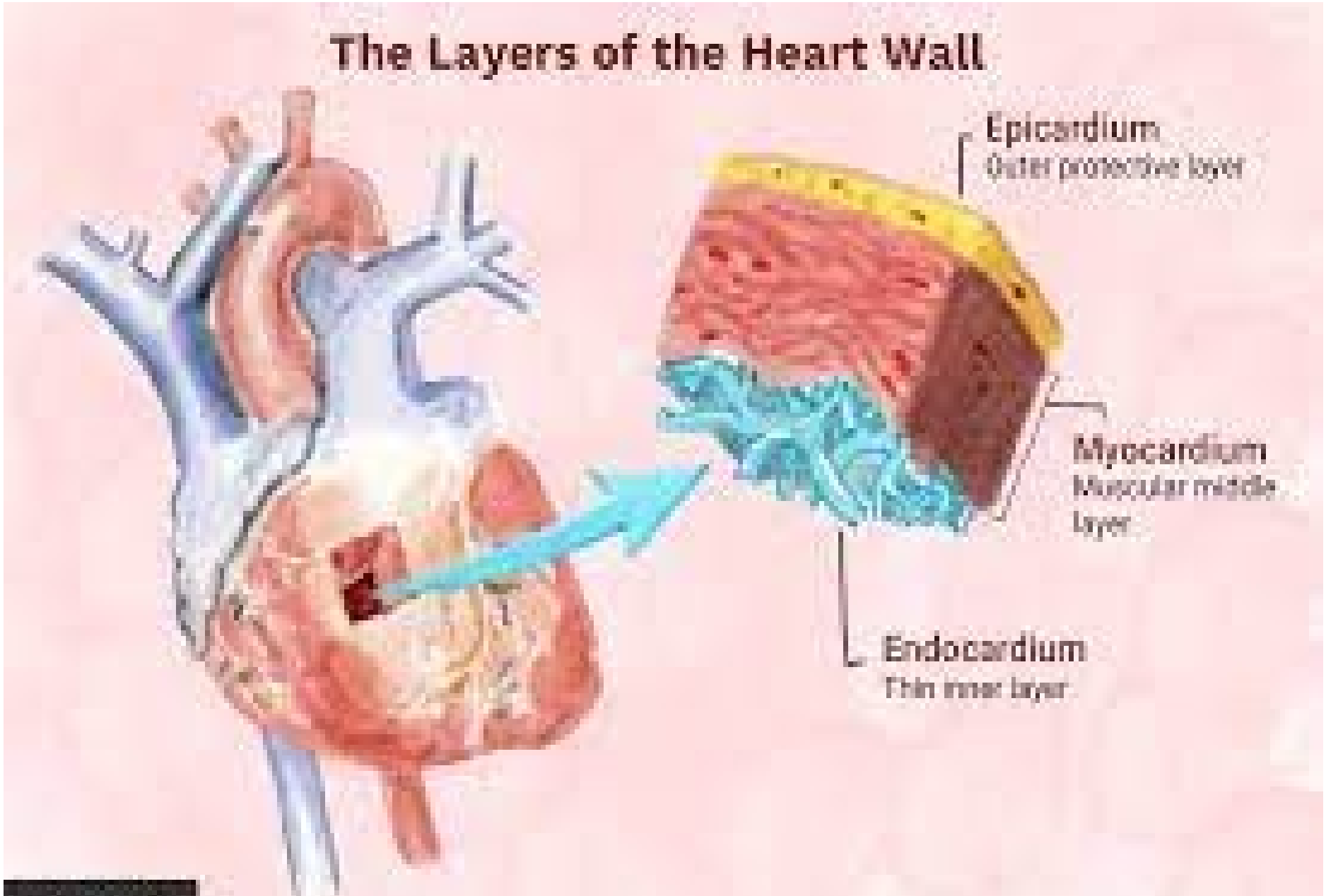
Coverings of the Heart:

- **Pericardium** – a double-walled sac that contains the heart and the roots of the great vessels.
  - A superficial **fibrous pericardium.**
  - A deep two-layer **serous pericardium:**

- The **Parietal Layer** Lines The Internal Surface Of The Fibrous Pericardium

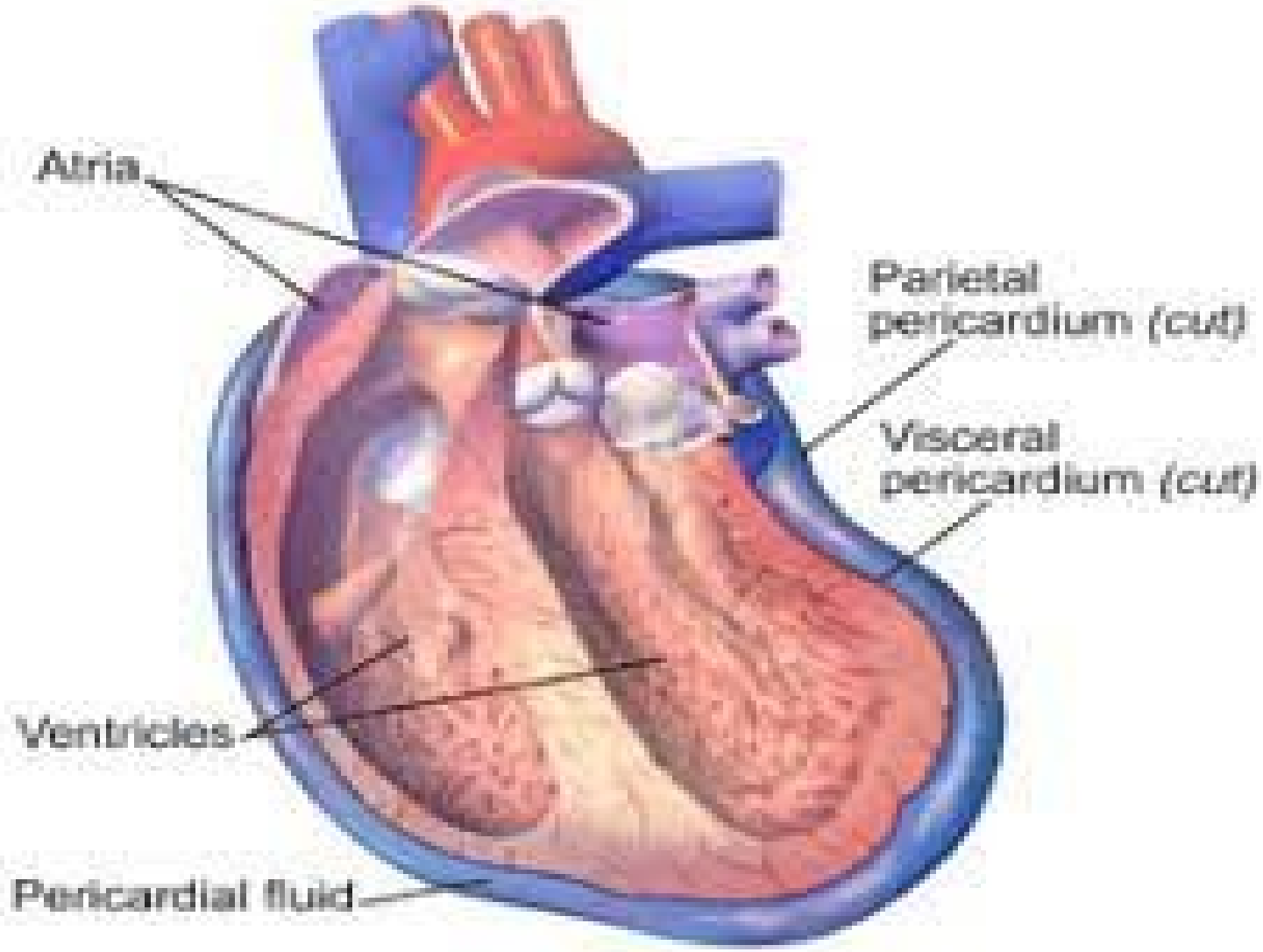
- The **Visceral Layer Or Epicardium** Lines The Surface Of The Heart

- They Are Separated By The **Fluid-filled Pericardial Cavity.**


The Layers of the Heart Wall
Epicardium
Outer protective layer
Myocardium
Muscular middle layer
Endocardium
Thin inner layer

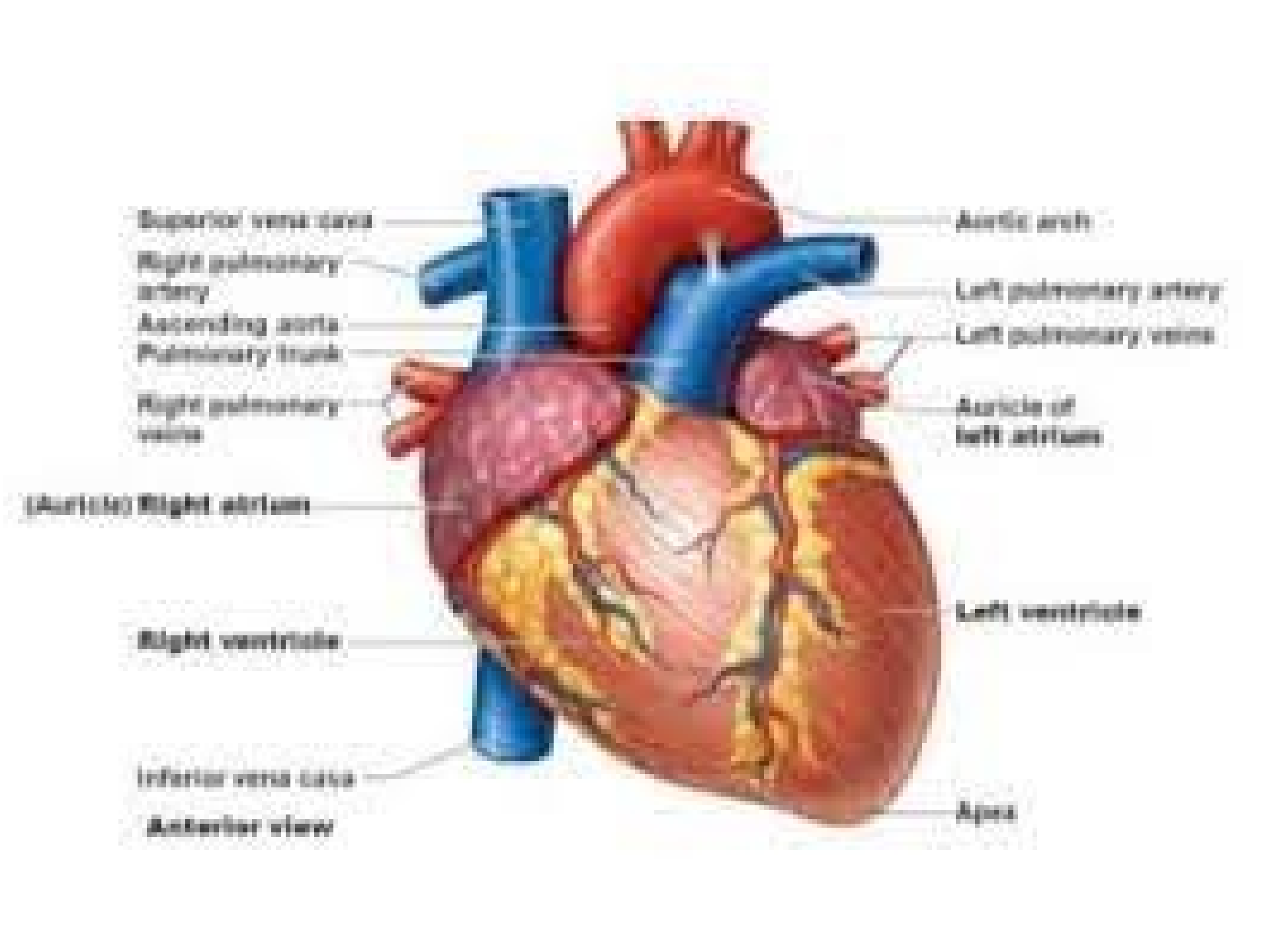
Coverings of the Heart:

- **Pericardium** – a double-walled sac that contains the heart and the roots of the great vessels.
  - A superficial **fibrous pericardium.**
  - A deep two-layer **serous pericardium:**
-
  - The **parietal layer** lines the internal surface of the fibrous pericardium
  - The **visceral layer or epicardium** lines the surface of the heart
  - They are separated by the **fluid-filled pericardial cavity.**



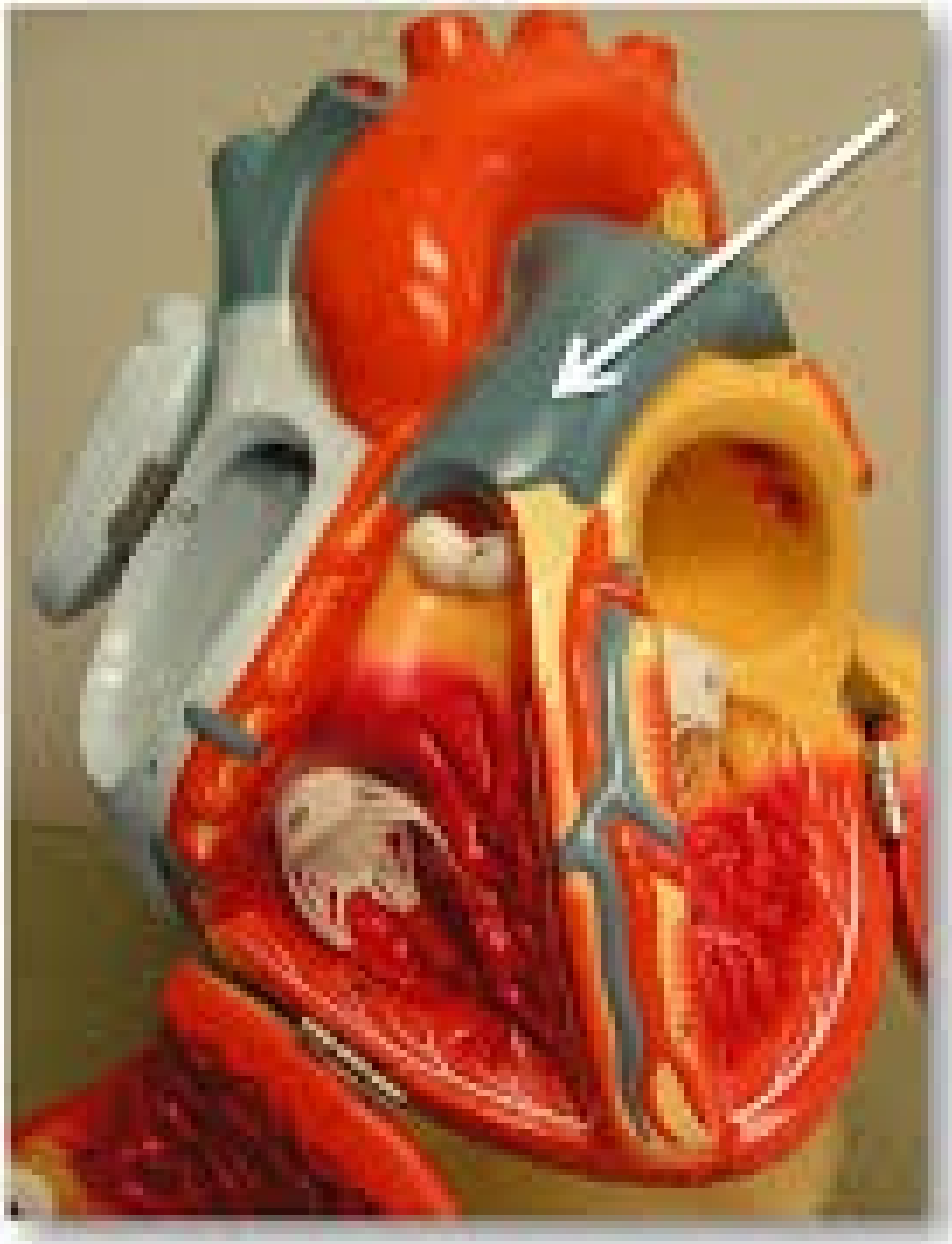
Pericardial Sac

- **Vessels** conveying blood away from the heart include:
  - **Pulmonary trunk**, which splits into right and left pulmonary arteries.
  - **Ascending aorta** (three branches) – brachiocephalic, left common carotid, and subclavian arteries.


Superior vena cava
Right pulmonary artery
Ascending aorta
Pulmonary trunk
Right pulmonary veins
(Auricle) Right atrium
Right ventricle
Inferior vena cava
Anterior view
Aortic arch
Left pulmonary artery
Left pulmonary veins
Auricle of left atrium
Left ventricle
Apex

Vessels that Supply/Drain the Heart (Anterior View):

- **Arteries –** right and left coronary (in atrioventricular groove), marginal, circumflex, and anterior interventricular arteries.
- **Veins –** small cardiac, anterior cardiac and great cardiac veins.

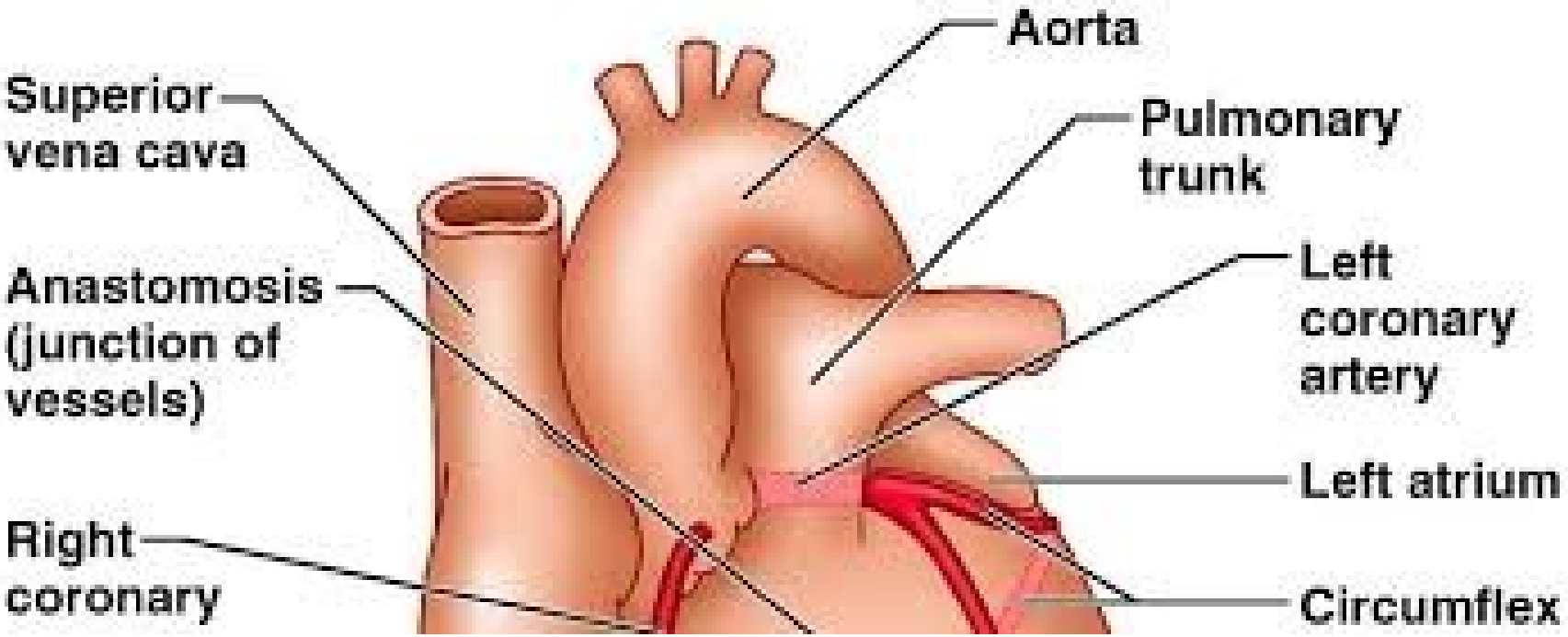


Pulmonary Trunk

Major Vessels of the Heart (Posterior View)

-
- **Vessels** returning **blood** to the heart include:
  - **Right and left pulmonary veins**
  - **Superior and inferior venae cavae**

- Vessels conveying blood away from the heart include:
  - **Aorta**
  - **Right and left pulmonary arteries.**


Aorta
Superior vena cava
Pulmonary trunk
Anastomosis (junction of vessels)
Left coronary artery
Left atrium
Right coronary
Circumflex

Vessels that Supply/Drain the Heart (Posterior View):

- **Arteries –** right coronary artery (in atrioventricular groove) and the posterior interventricular artery (in interventricular groove)
- **Veins** – great cardiac vein, posterior vein to left ventricle, coronary sinus, and middle cardiac vein.

THANK YOU